EUROPEAN MECHANICS SOCIETY

Sparse-sensing state estimation of vortex-gust airfoil interactions

Toward model-based flow control

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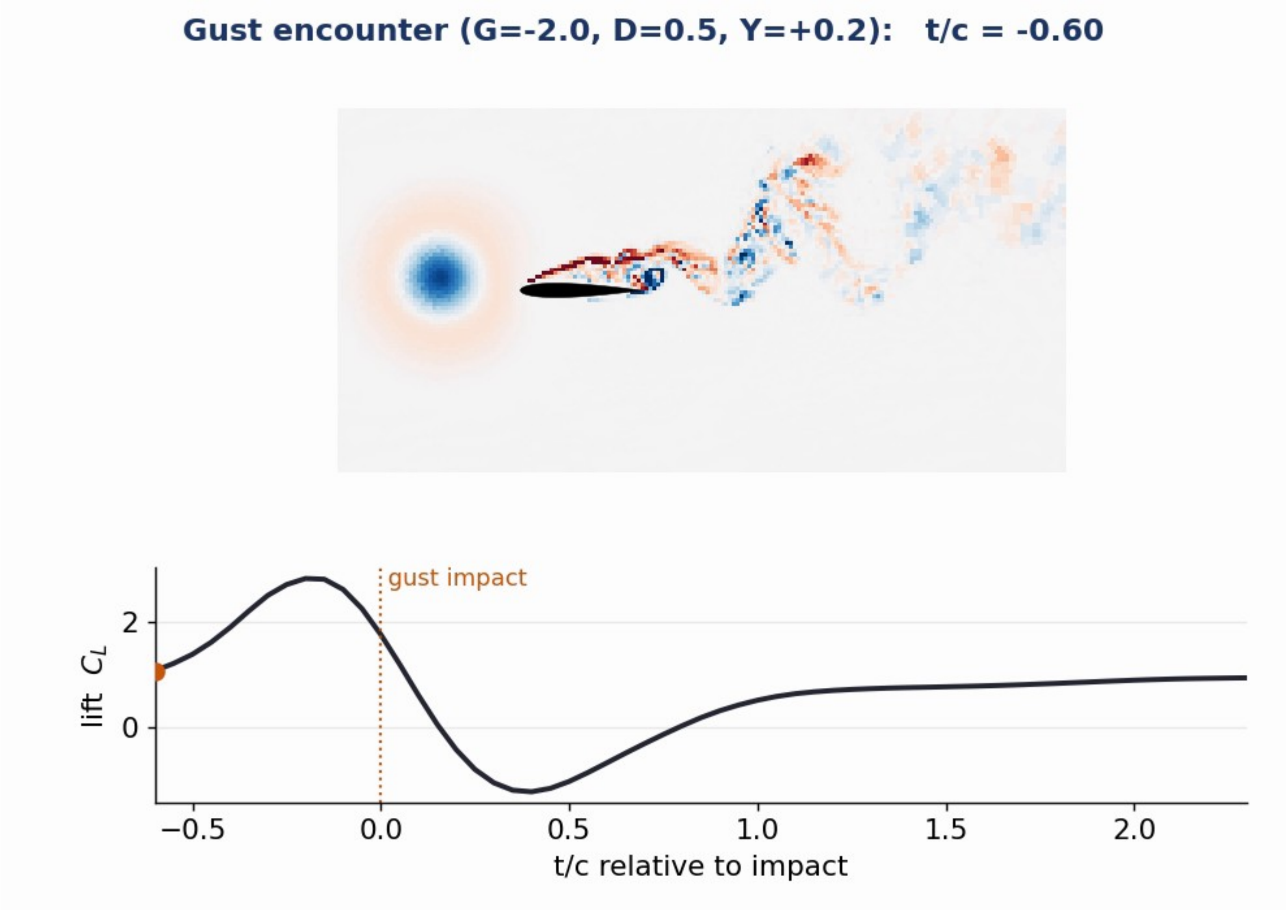
EUROMECH Colloquium · Data-driven active control in flows: from model-based to reinforcement learning

The control problem: gusts are fast, strong, nonlinear

- Small and micro air vehicles increasingly fly where the **gust speed is comparable to or above the flight speed** (urban canyons, building and ship wakes): gust ratio **G > 1** brings large, fast load transients classical models miss
- A vortex strikes the wing: a **leading-edge vortex** forms and sheds, and the lift swings strongly (right: DNS, a moderate **G = -2** gust, C_L from about -1 to +3)
- Active control, model-based or RL, needs a **fast, faithful forward model** it can trust under rollout
- **This talk: what reduced state is worth planning against?**



Objective: a reduced state that keeps the wake

- The load transient is built by the **leading-edge vortex (LEV)** and the wake it leaves: the discriminating information is in the **wake**, not the integrated forces C_L, C_D
- We quantify it by the **large-scale wake enstrophy**: a Gaussian scale split (Motoori & Goto 2019) at $\sigma/c = 0.05$ isolates the load-bearing large-scale vorticity (LEV + shear layer) from fine turbulence, then integrates ω_L^2 over the wake
- **Objective:** find a reduced state that faithfully **encodes the wake** (the part reconstruction loses), not just the forces, and that is **recoverable from sparse sensors**
- And **compare candidate states**, predictive vs reconstructive vs linear, under one matched protocol

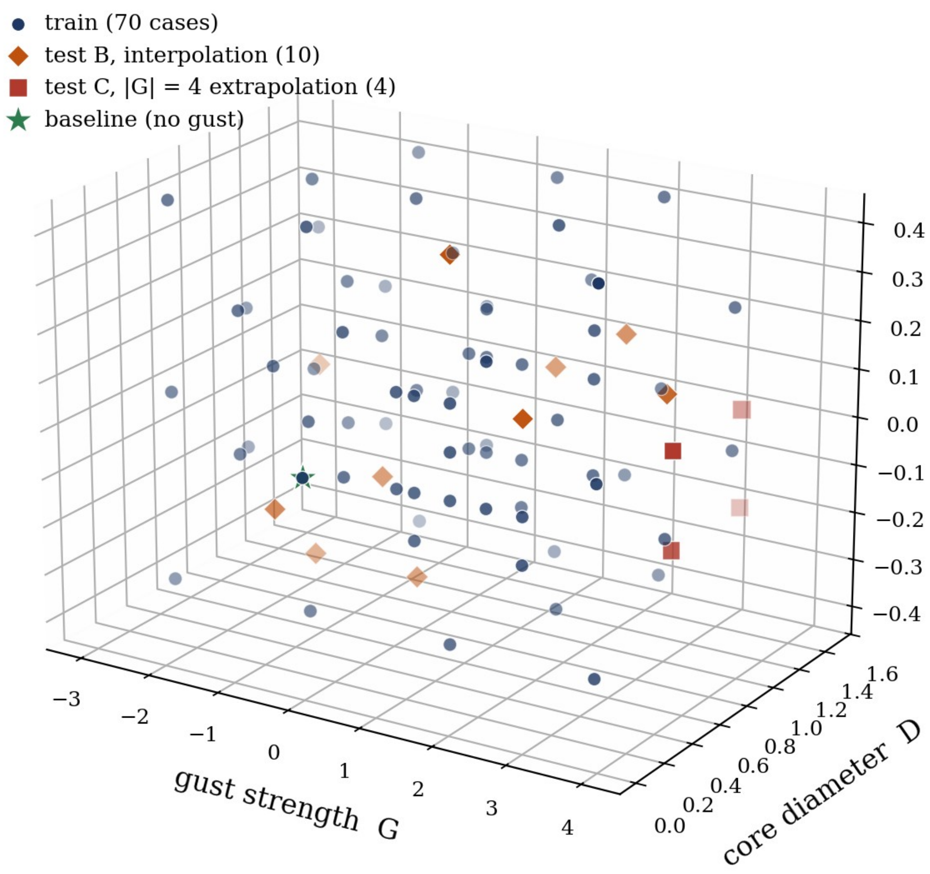
The opening for JEPA

- Most flow ROMs, POD/DMD, autoencoders, are trained to **reconstruct the field**
  - reconstruction fixes the latent only up to a diffeomorphism, so its geometry is not constrained to be predictable
- Alternative: learn a state that is **predictive by construction**, and plan against it

Which reduced state faithfully encodes the wake that reconstruction loses, and is recoverable from sparse sensors?

Data

- **DNS** (SOD2D, no subgrid-scale model), NACA 0012, α = 14°, **Re = 5000**
- Taylor-vortex gusts parametrised by strength **G**, core diameter **D**, wall-normal offset **Y** (right: the 84-case envelope, isometric)
- Impact-centred encounters; six observables: C_L, C_D, impulse I_y, **wake enstrophy**, ±circulation
- Held-out **test_b** (interpolation) and **test_c** (|G| = 4 extrapolation)



Training envelope in (G, D, Y): train, test_b interpolation, test_c |G| = 4 extrapolation, baseline.

What is a JEPA? (1 / 2)

Autoencoder (most flow ROMs): reconstruct the field

field ω(t) → Encoder → latent z → Decoder → reconstructed field

loss compares the reconstruction to the input in PIXEL space

JEPA: predict the next latent, no decoder

field ω(t) → Encoder → latent z(t) → Predictor → predicted z(t+1)

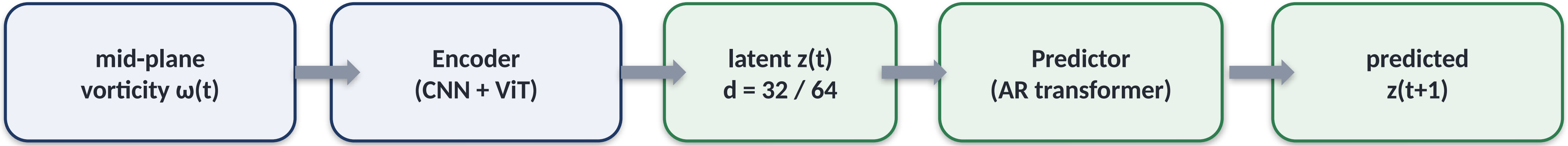
loss compares predicted z(t+1) to z(t+1) = Encoder(next frame) in LATENT space; an anti-collapse term replaces the decoder

Reconstruction ties the latent to pixels; prediction makes it forecastable, which is what a controller needs.

What is a JEPA? (2 / 2): the recipe and why for control

- **Training, in one line:** minimise the **latent** prediction error plus an **anti-collapse term** so the encoder cannot cheat by mapping everything to a constant (no decoder, no pixel loss)
- So the encoder **keeps only what predicts the future** (the control-relevant dynamics), and the predictor **gets no pixel-level shortcuts**
- **Cheap to roll out:** planning happens in the $d = 32$-64 latent, two to three orders of magnitude cheaper than evolving the full field
- Lineage: I-JEPA / V-JEPA (image, video); from-pixels LeWM, LeJEPA, PLDM, so far only gridworld and toy visual tasks
- **Why for control:** a learned world-model that is **observable from sensors**; to our knowledge the **first end-to-end JEPA on a parametric flow**

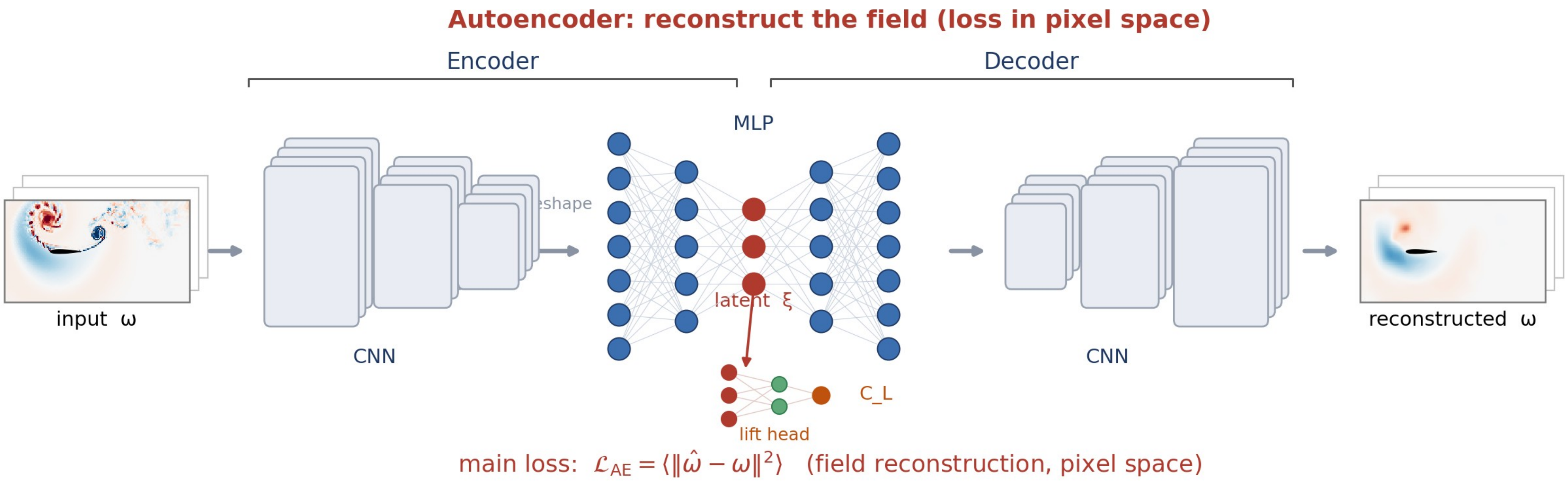
Our architecture for gust encounters



- The **encoder** maps the field to a latent: a pure state map $\omega(t) \to z$
- The **predictor** advances the latent $z(t) \to z(t+1)$ from the latent history; it **is** the latent dynamics model you plan or learn against
- A **visualisation decoder** is trained on the **frozen** encoder, never in the loss (figures only)
- Compared at matched latent dimension against a reconstructive AE (Fukami) and POD

Illustrated alternative: the autoencoder route

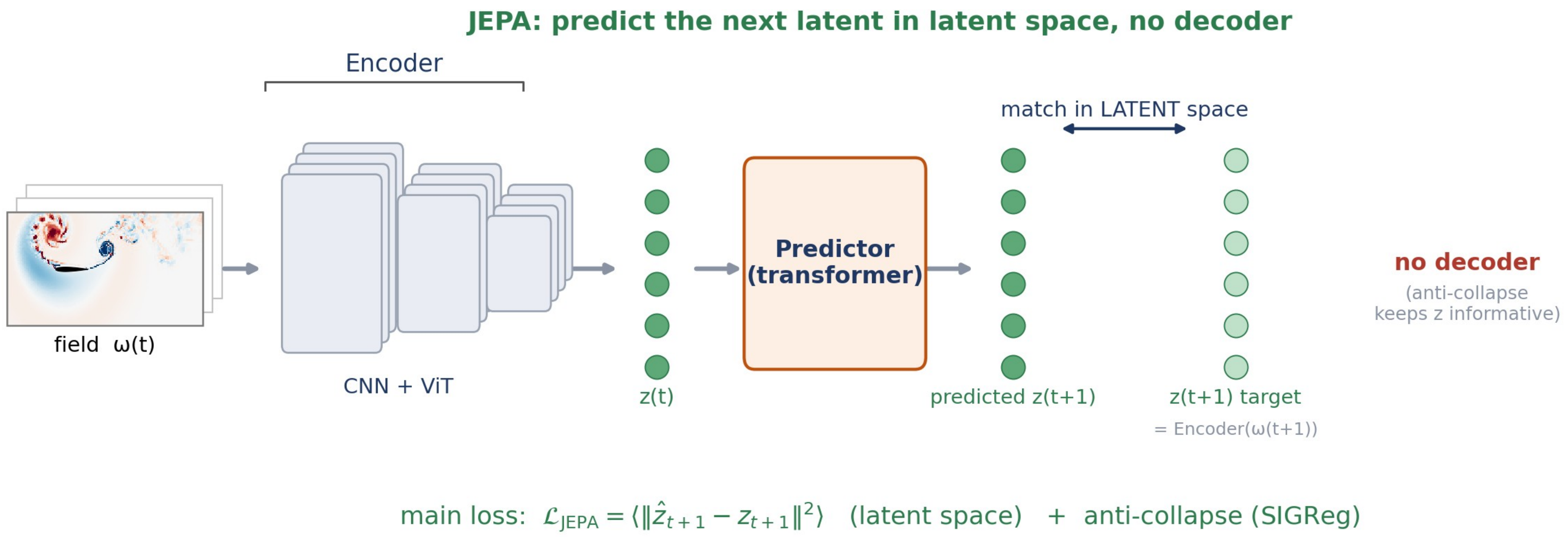
- Reconstructive route (Fukami-Taira lineage): a **CNN encoder**, an **MLP latent ξ** (which also feeds a lift head to C_L), and a **CNN decoder**; the loss compares the reconstructed field to the input in **pixel space**



An autoencoder reconstructs the field; the latent is shaped by pixel reconstruction.

Illustrated alternative: the JEPA route

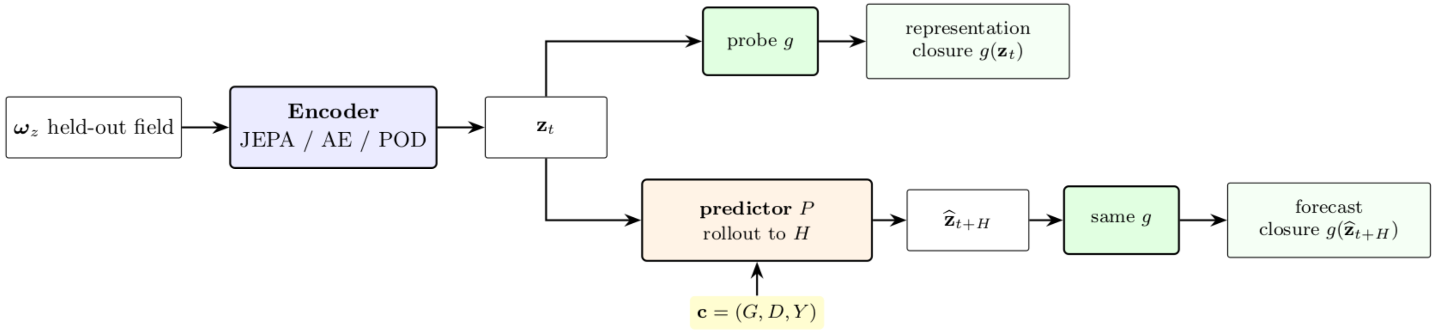
- Same encoder, but a **predictor** advances the latent and the loss lives in **latent space**, with **no decoder** (an anti-collapse term keeps z informative)



JEPA predicts the next latent in latent space, with no field reconstruction.

Closure protocol

- **Representational closure (the headline):** probe each observable from the **encoded** latent at impact + 16, what the state *carries*, no rollout
- **Forward (rollout) mode:** roll the predictor recursively to **H = 16** and probe the predicted latent; here we use it **qualitatively** (does the rolled latent track the wake?)
- **Matched protocol:** same predictor architecture and same probe family, **trained / fitted separately per latent**, so differences are the encoder's
- **Parameter-only floor:** can the **three gust numbers (G, D, Y) alone** predict each observable (a kernel-ridge fit)? Beating the floor proves the latent **carries flow state beyond the parameters**
- Reported with boot



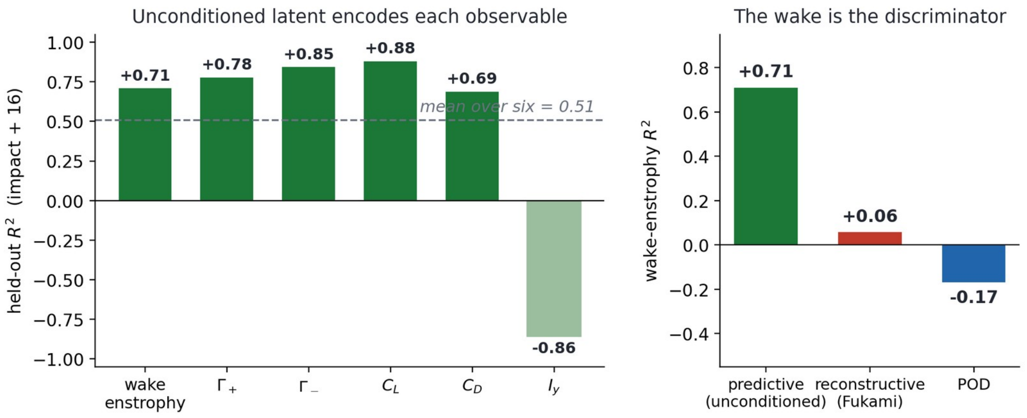
The predictor architecture, the conditioning $\mathbf{c}$, and the probe family g are identical across the three encoder families, each trained or fitted separately per family.

The same probe maps every encoder family to the observables. Representational mode (encoded latent) is the headline; forward rollout is shown qualitatively.

RESULT

Main result: the latent keeps the wake

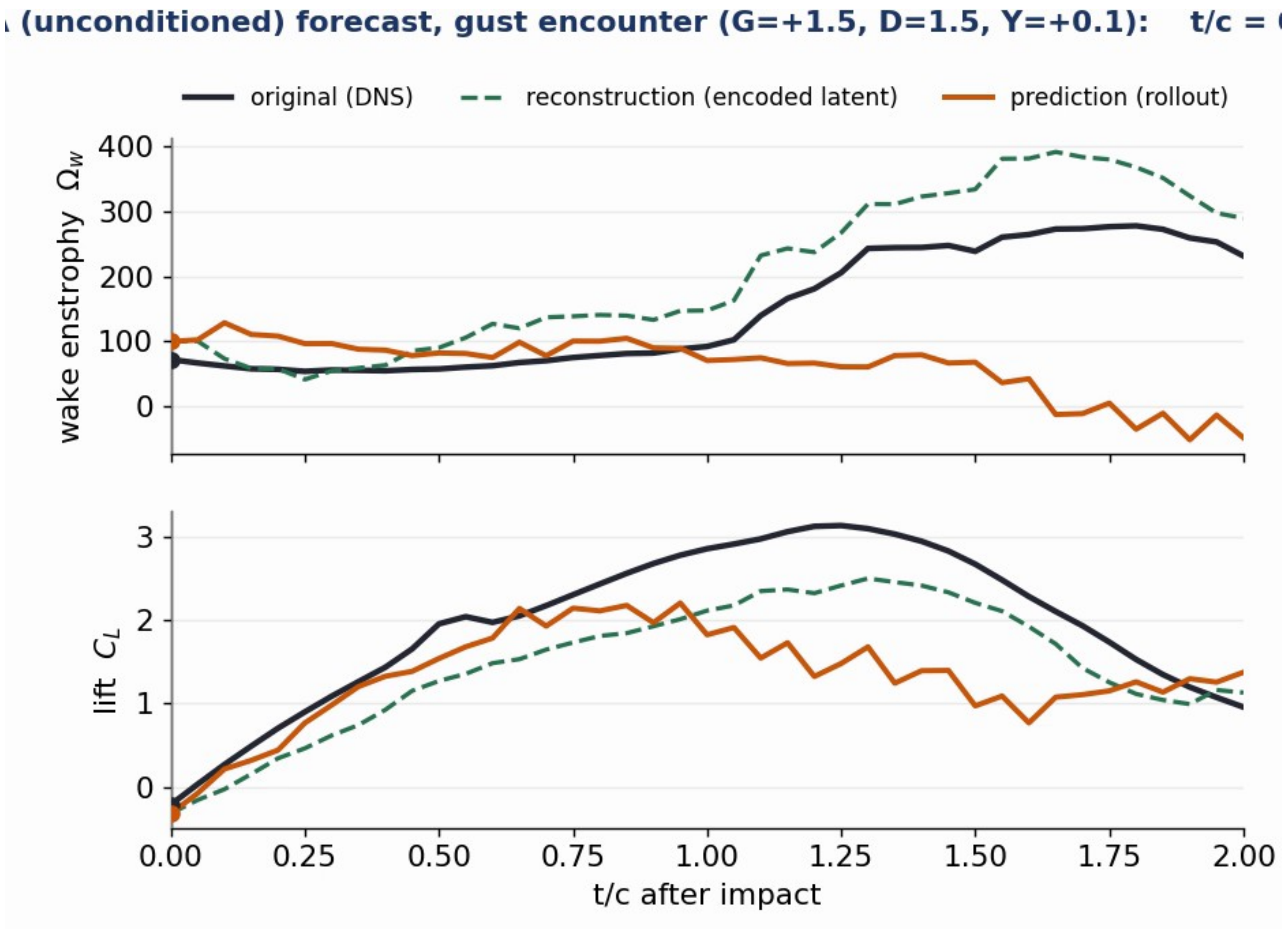
- **Representational closure:** probe the **encoded** latent (no rollout) for each observable at impact + 16; the question is what the unconditioned state *carries*
- **Wake enstrophy is the discriminator: tf-no-c R^2 = 0.71** (the conditioned model 0.75; reconstructive and POD far lower, 0.06 and below)
- So the **unconditioned** latent (no gust parameters anywhere) **encodes the wake that reconstruction smooths away**
- Forces and circulations are read off cleanly too (C_L 0.88, Γ± 0.85 / 0.78, C_D 0.69); the **mean over six is 0.51**, with the transport impulse I_y the weak observable
- **Wake enstrophy** E_w = ∫ ω_z² over the wake (x/c ∈ [0.5, 4], |y/c| ≤ 1): the **intensity of the wake's rotational structures** (LEV + shed vortices) that set the future load, the part reconstruction-only states lose



Held-out representational R^2 at impact + 16 from the encoded latent (test_b). Left: per observable (tf-no-c). Right: wake-enstrophy R^2 across families.

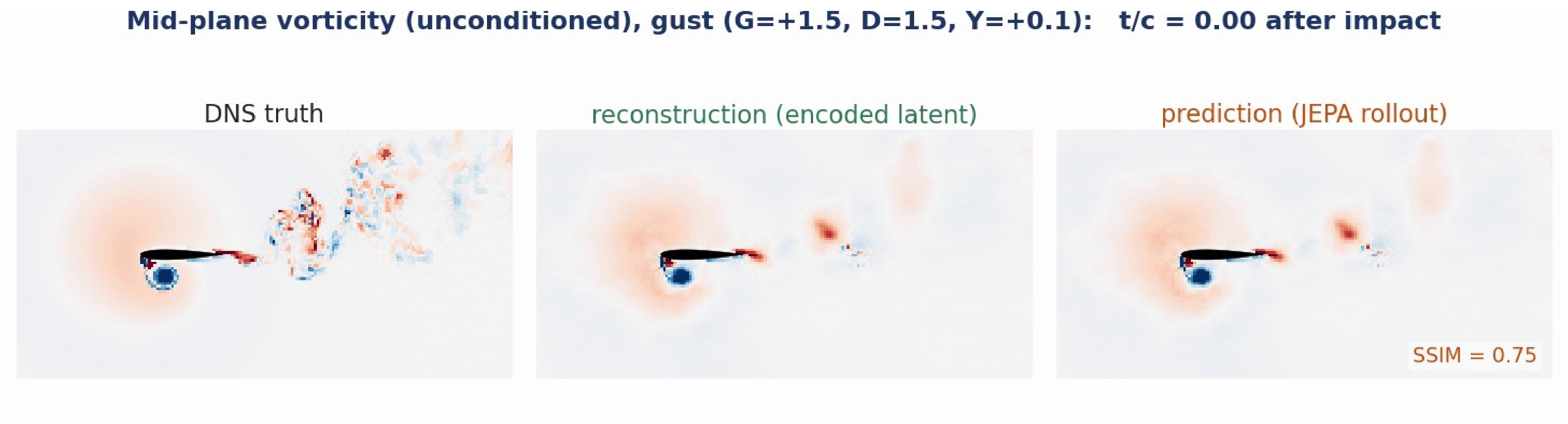
The rolled latent qualitatively tracks the wake

- The predictor **rolls the latent forward from impact**; each scalar is read off the rolled latent by a fixed linear probe
- **Wake enstrophy** and **lift** qualitatively track the DNS truth (black) through impact and the lift dip
- Reconstruction (green) = encoded-latent **ceiling**; prediction (orange) = the **rolled latent**
- A representative **low-error** encounter (G = +1.5), not the hardest case; shown qualitatively, no closure R^2 claimed here



The same rollout in physical space

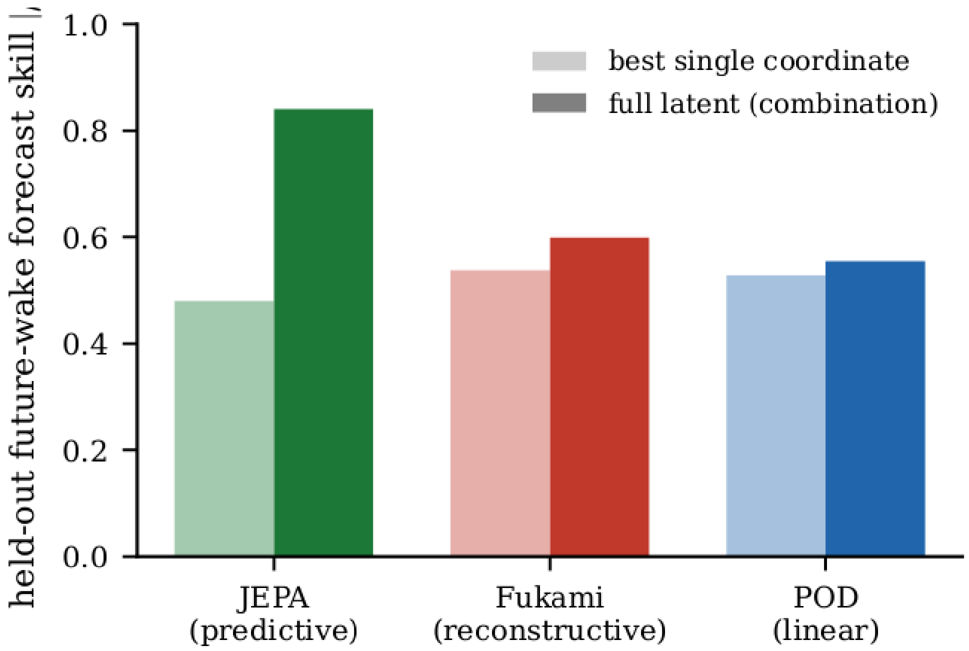
Decode the rolled latent every frame: the rolled latent qualitatively tracks the wake as full vorticity fields, not just scalars.



DNS truth vs reconstruction (JEPA encode→decode of the latent) vs prediction (rolled latent), G = +1.5. At impact (H = 0) prediction = reconstruction; they diverge with horizon. SSIM(prediction) 0.75 at impact, 0.63 at H = 16. The latent keeps the LEV and shear layer; fine-scale wake is not retained at d = 64.

It is the wake, not the forces

- **Forces** (C_L, C_D) are carried **redundantly by every family**, many coordinates encode them
- Only the unconditioned predictive latent carries a **distributed wake code** (full latent 0.84; best single coordinate 0.48)
- Reconstructive / linear: best single coordinate ≈ the whole latent → **no collective wake structure**
- And it clears the **parameter-only floor** → the latent, not the parameters, does the work

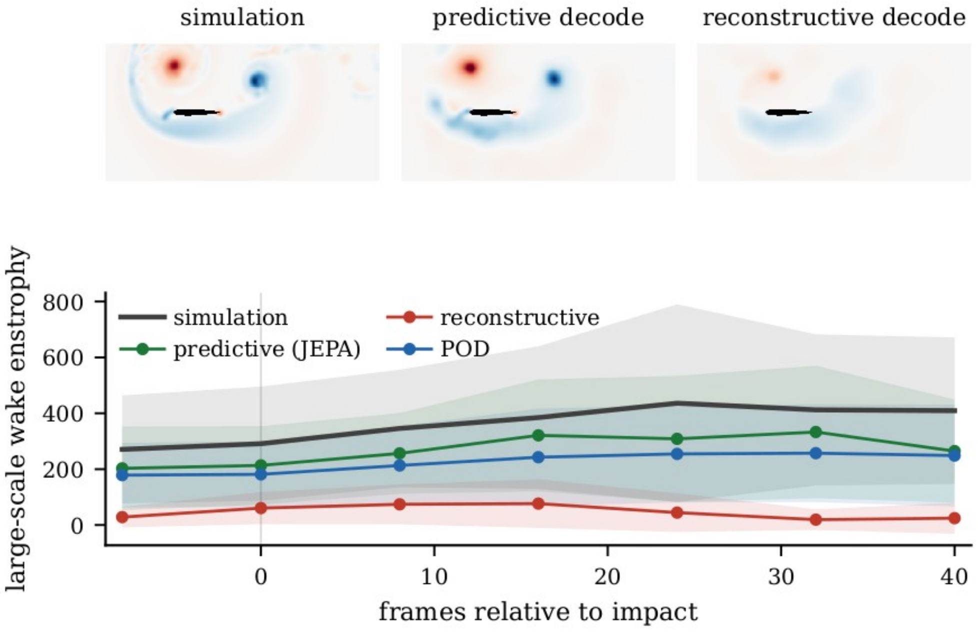


Wake skill: full latent (dark) vs best single coordinate (light).

RESULT

The wake in physical space: the Gaussian scale split

- **Gaussian scale split** (Motoori & Goto 2019): low-pass the vorticity at σ/c = 0.05 into a **large-scale** part (LEV + shear layer, carries the lift) and a fine part
- Top: large-scale vorticity at impact+16 for the strongest test gust (G = -3, D = 1.5, Y = -0.1), simulation vs the **predictive** and **reconstructive** encode-then-decode reconstructions: predictive keeps the LEV/wake, reconstructive smooths it
- Bottom: the **large-scale wake enstrophy** through the encounter; bands = mean ± 1 s.d. across test_b encounters
- So 'it is the wake' is visible in **physical space**, not only in the scalar closure

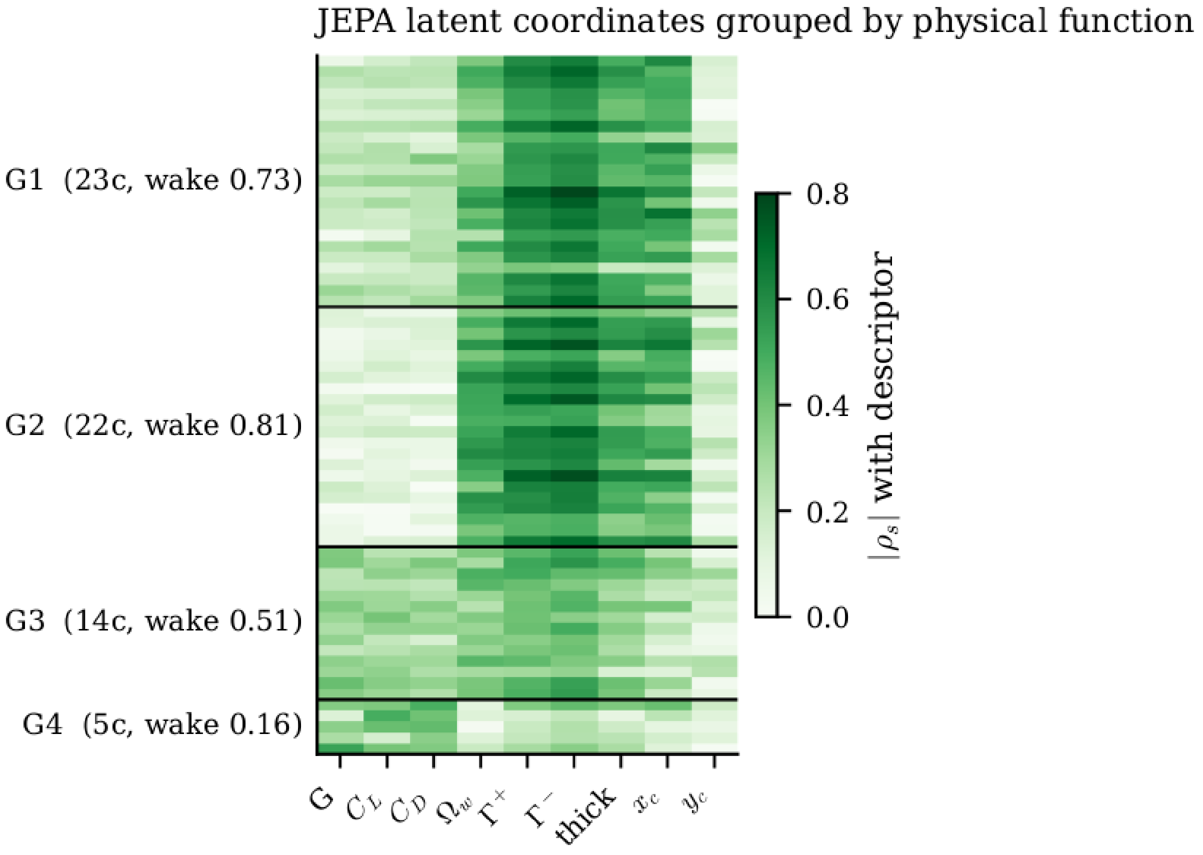


Large-scale (σ/c = 0.05) wake vorticity at impact+16 (top) and large-scale wake enstrophy vs frame (bottom), by family.

RESULT

Latent coordinates group by physical function

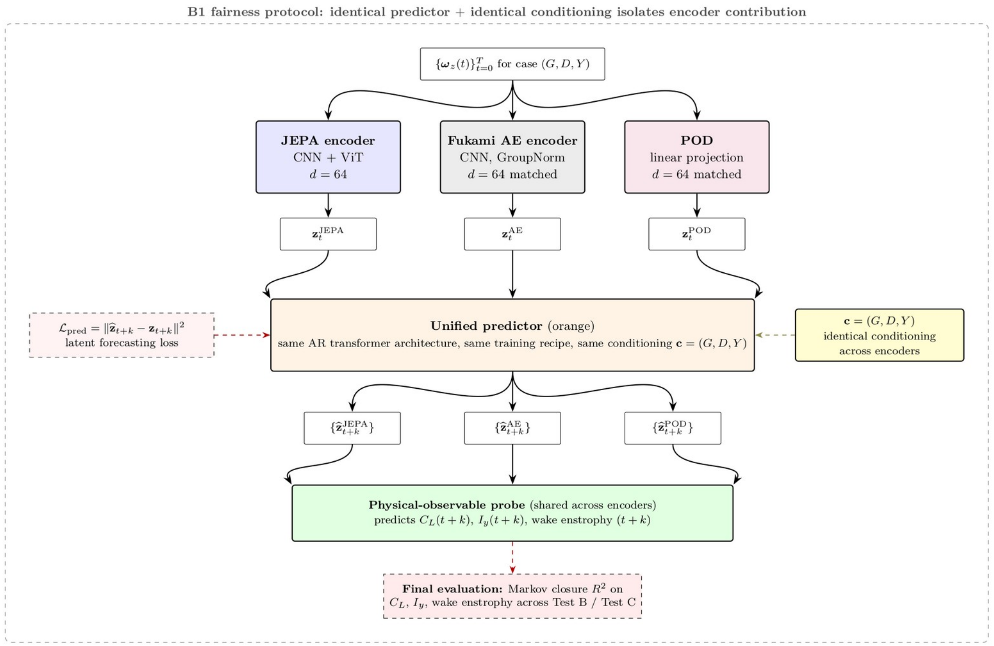
- Profile each of the 64 unconditioned-latent coordinates by its |Spearman| correlation with nine descriptors (gust G, forces, wake enstrophy, circulations Γ±, wake thickness, centroid)
- Clustering the profiles recovers **functional groups**: **~51 wake-vorticity** coordinates (three wake-dominated clusters) and **11 gust-forcing** coordinates (one force cluster)
- Alone, the wake groups carry the wake at **≈ 0.7-0.8**, the forcing group at **0.45** (all 64 together: 0.84), the collective wake code seen earlier
- Read **descriptively**, not as a causal claim



Per-coordinate |ρ| with each descriptor; rows grouped into functional clusters (G1-G4), labelled with coord count and held-out wake skill.

Controls: objective and supervision, not architecture

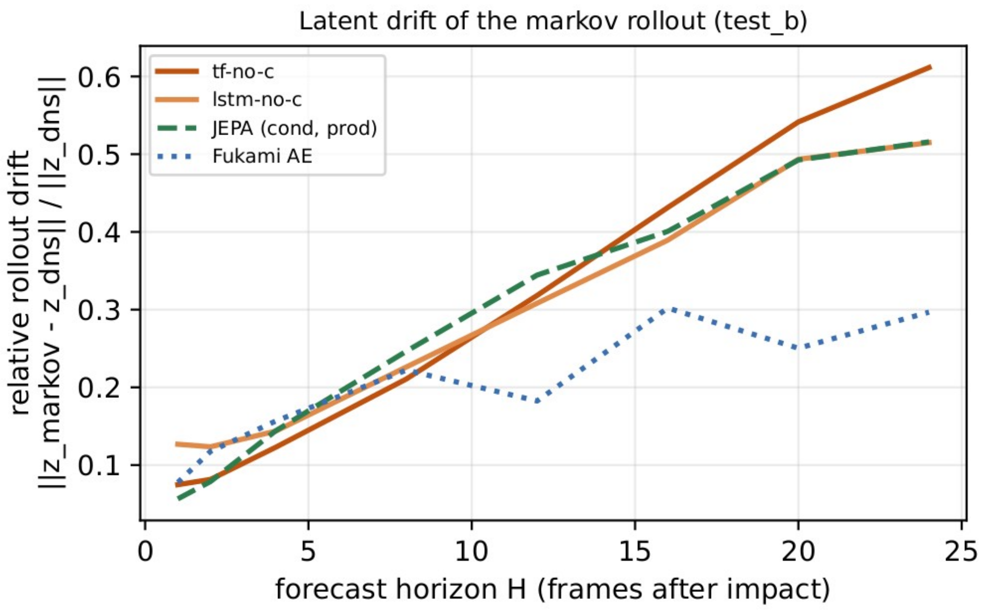
- **(Conditioned-model control; no unconditioned variant run.) 2 × 2:** objective {predictive, reconstructive} × architecture {CNN, CNN+ViT}, auxiliary heads matched
- Predictive beats reconstructive at **both** architectures (wake R^2 0.46 / 0.45 vs 0.16 / 0.29) → **not the ViT**
- Remove the wake head from the predictive model → wake closure **collapses below floor (R^2 = −1.03)**
- → the gain needs the **predictive objective and wake supervision together**



Objective × architecture controls, three seeds per cell.

Diagnostic 1: latent drift

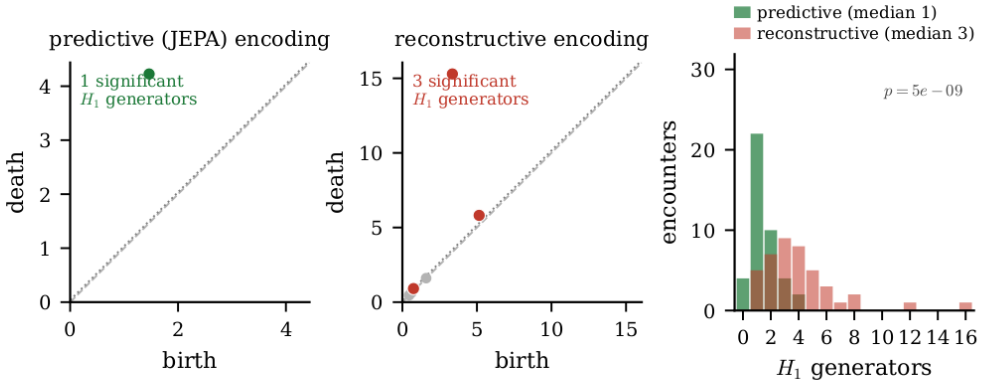
- **Why this question:** a planner / RL agent queries the model at states reached by its **own rollout**, one-step error is not enough
- **Mahalanobis distance:** covariance-aware distance from a reference cloud, d = sqrt((z−μ)' S^-1 (z−μ)); ≈ 1 is one standard-deviation unit
- We compare the **rolled-out** latent to the distribution of **DNS-encoded** latents (ratio = rollout / encoded)
- **Result:** the **unconditioned** rollout drift grows **gracefully** and stays on-manifold, matching the conditioned model; the reconstructive rollout drifts far off-manifold by comparison
- So the predictive state remains in-distribution as it is propagated, exactly where the probes (and a planner) are valid



Rollout drift vs horizon (test_b): relative Mahalanobis distance of the rolled latent to the DNS-encoded cloud. The unconditioned rollout stays on-manifold.

Diagnostic 2: topology of the encounter

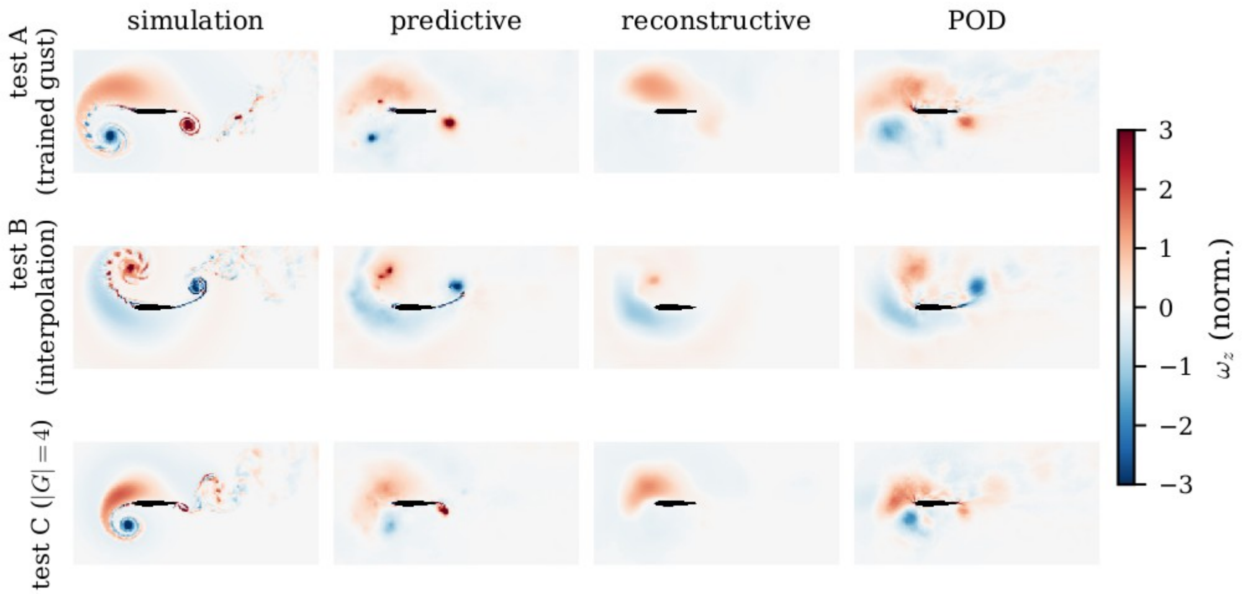
- **Why:** shedding and gust encounters are **cyclic**; a faithful state should trace **one loop** per encounter, not a tangle
- **Persistent homology** (coordinate-free): it grows a distance scale on the latent trajectory and counts the **loops (1-cycles) that persist** over a long scale range, i.e. the genuine cycles, not noise
- **Result (encoded latent, 42 test_b encounters):** the **unconditioned** tf-no-c latent has **median 1** persistent loop vs reconstructive **median ~4** (Mann-Whitney $p \approx 5\times10^{-9}$)
- So the predictive state is a **single clean cycle**, the reconstructive one **fragments**. This is the *topological count*, **not** a claim that the PCA curve visually closes (panels are encode→decode, no predictor; the orbit departs the baseline cycle and only partially returns in 120 frames)



(a, b) PCA of the encoded-latent trajectory; (c) staged decodes (reconstructions) with OT distances. Single-cycle = median 1 persistent H1 generator (vs ~4); the orbit need not visually close in the window.

Decoded reconstructions

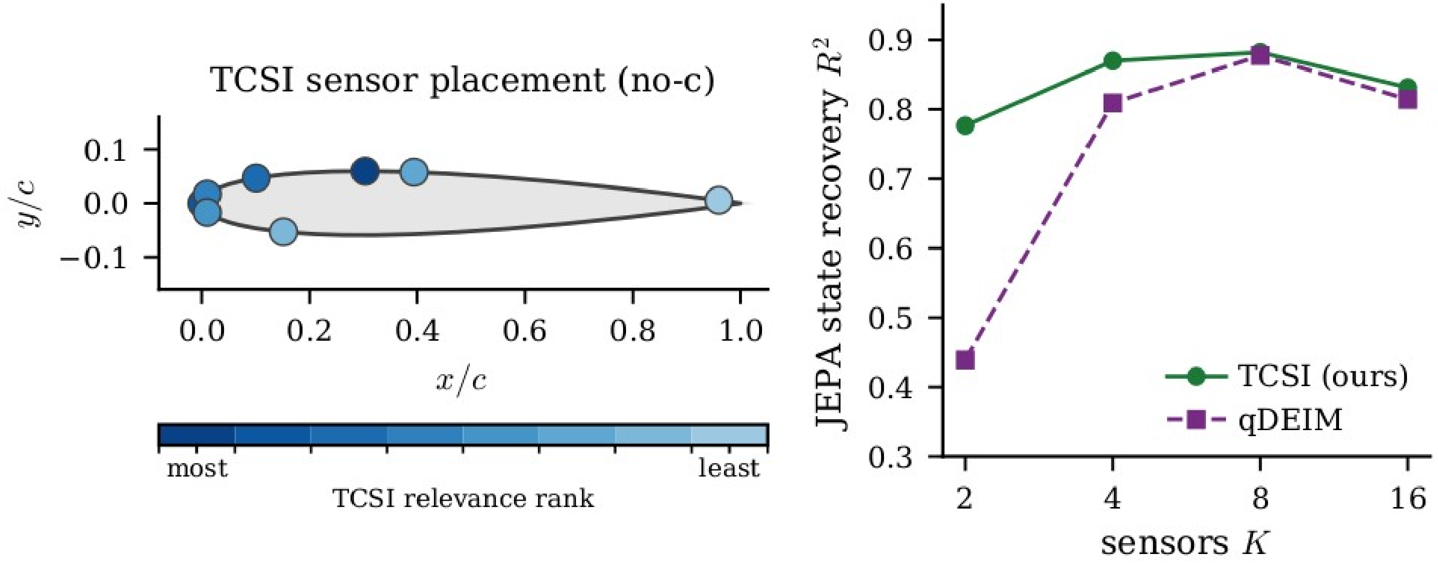
- Decoded fields: predictive vs reconstructive vs POD vs DNS
- The predictive decode is **blurrier pixel-wise** but preserves the **transported large-scale wake**; the reconstructive decode is sharp yet drifts under rollout



Held-out reconstructions; the predictive objective trades pixel sharpness for transported structure.

Sparse sensor placement

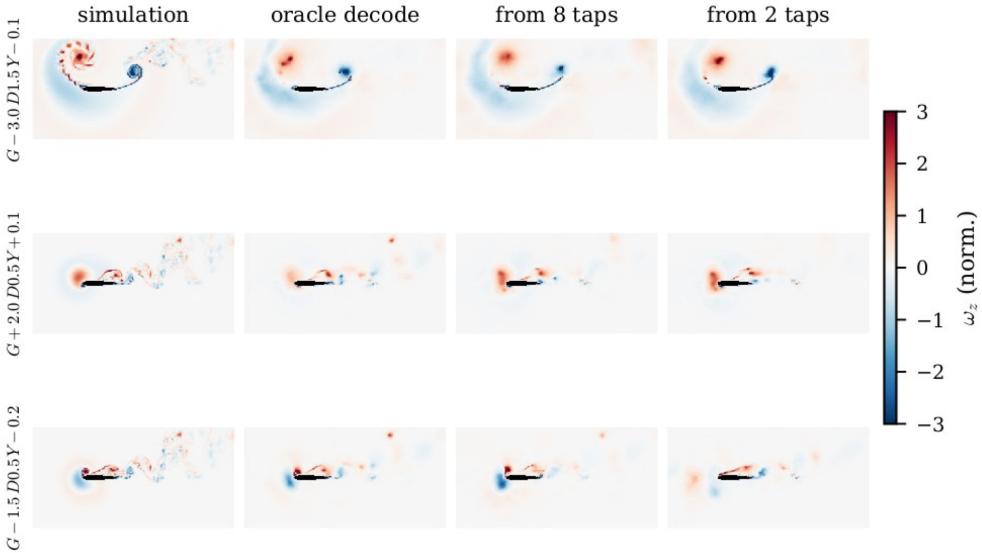
- Sparse wall-pressure sensors selected by **TCSI / qDEIM** vs uniform
- Pressure → latent map: KernelRidge(RBF) on the K-sensor × pre-impact-window vector
- K = 2 / 4 / 8 / 16; the LSTM/KRR comparison is 5-fold-CV selected to guard small-sample overfitting



Optimal sparse sensor placement on the airfoil surface.

Flow recovered from sparse wall pressure

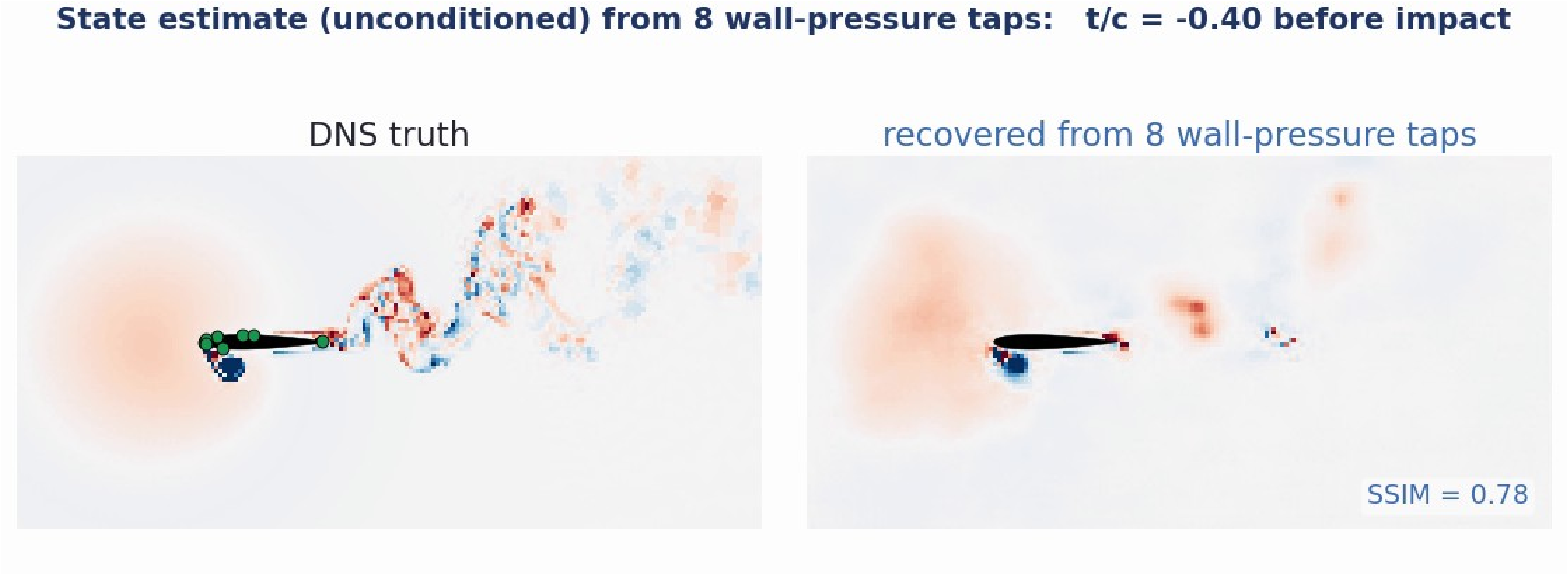
- **Model:** a **KernelRidge (RBF kernel)** regressor maps the K wall-pressure taps over a pre-impact window to the **impact-frame latent**; the frozen decoder then renders the field
- **K = 8 taps** recover the leading-edge vortex and shear layer; **K = 2** coarsens but keeps the gross wake structure
- Benchmarked against the **oracle decode** (decode of the simulation-encoded latent) and the simulation
- So the predictive state is **reconstructible from a few wall sensors**, a deployment-relevant observability result



Flow recovered from sparse wall pressure: simulation, oracle decode, and decode of the latent estimated from K = 8 and K = 2 taps (held-out encounters).

Sparse-sensor state estimation in action

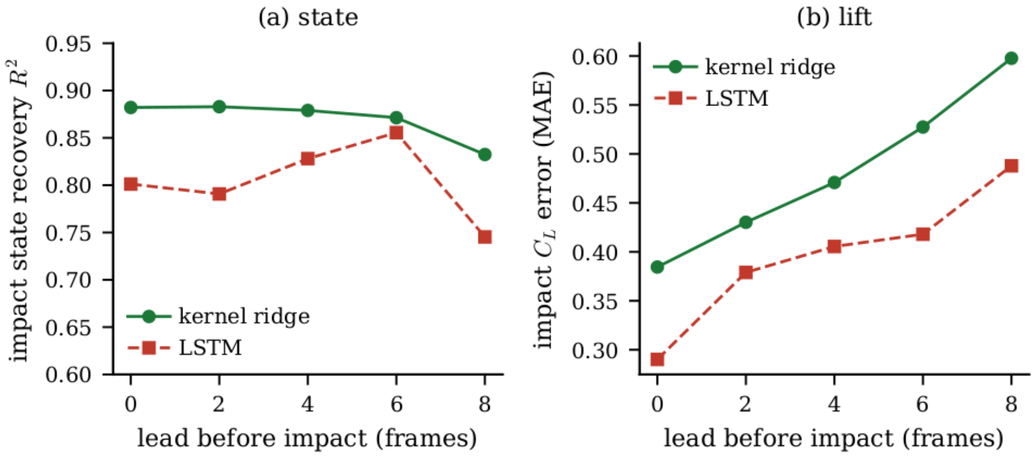
No predictor here: a per-frame **MLP** maps a causal 6-frame window of 8 wall-pressure taps to the latent, then the frozen decoder renders the field.



DNS truth (green dots mark the 8 TCSI taps) vs the field recovered from wall pressure alone, on a held-out encounter. Held-out-encounter latent R^2 = 0.74, SSIM 0.63. Wall pressure fixes the near-body LEV and shear layer; the far wake is not surface-observable.

Control relevance: observable ahead of impact

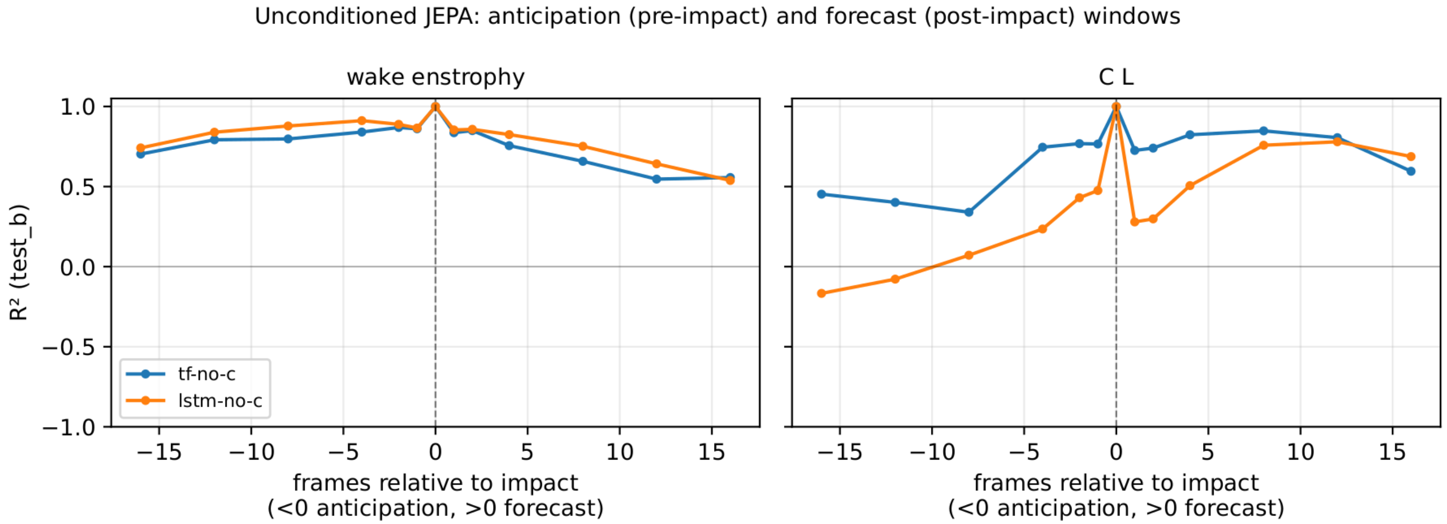
- From a causal window of sparse wall pressure, recover the **impact-frame latent** at a **lead time before impact** (held-out test_b)
- **Impact state** is recovered at **R^2 ≈ 0.88** right at impact and stays **above 0.83 out to 8 frames ahead** (kernel ridge; LSTM comparable)
- **Impact lift** C_L is estimated with MAE rising gracefully from **≈ 0.38 at impact to ≈ 0.60 at 8 frames ahead** (LSTM lower, 0.29 to 0.49)
- So the **unconditioned** state is **observable ahead of impact** from a few wall sensors: the ingredient a controller consumes



Recovered from sparse wall pressure at each lead time before impact (test_b): (a) impact-state R^2; (b) impact-C_L MAE. Kernel ridge vs LSTM.

Two forecast windows: early warning, then forecast

- **Roll the unconditioned predictor** and read each observable off the rolled latent: how early can we **anticipate** impact, and how far can we **forecast** after it?
- **Wake:** a wide **±16-frame (≈ ±0.8 t/c)** window around impact (R^2 0.7-0.9) -- the structural signature gives **early warning** and short-range forecast
- **Lift C_L:** hard to anticipate (flat before impact) but **strongly forecastable after** (R^2 ≈ 0.8 to +12) -- the load is predictable **once impact hits**



Wake (left) and lift (right) R^2 vs frames relative to impact (test_b): x < 0 anticipation lead, x > 0 forecast horizon; tf-no-c and lstm-no-c.

Conclusions

- A **fully unconditioned predictive latent** (no gust parameters anywhere) **representationally keeps the wake** (wake $R^2 \approx 0.71$) that **force-only and reconstruction-only** states lose
- That latent **traces a single clean cycle**, is **transport-consistent** within an encounter, and is **observable from sparse wall pressure**
  - **compact** ($d = 32 \approx d = 64$, participation ratio ≈ 1.7)
- The **rollout** is shown **qualitatively**: it stays on-manifold and tracks the wake, but we make **no forward-closure R^2 claim**
- It is a **substrate** for model-based control and RL world-models, **not yet a validated controller**
- **Next:** add an **actuation channel** (same machinery), **close the loop**, push to **3D observability** at $|G| = 4$, and carry the recipe to **other parametric flows**

Acknowledgements

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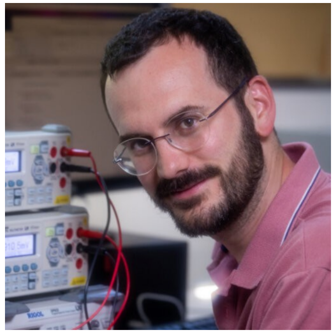
Alberto Solera-Rico

INTA · UC3M



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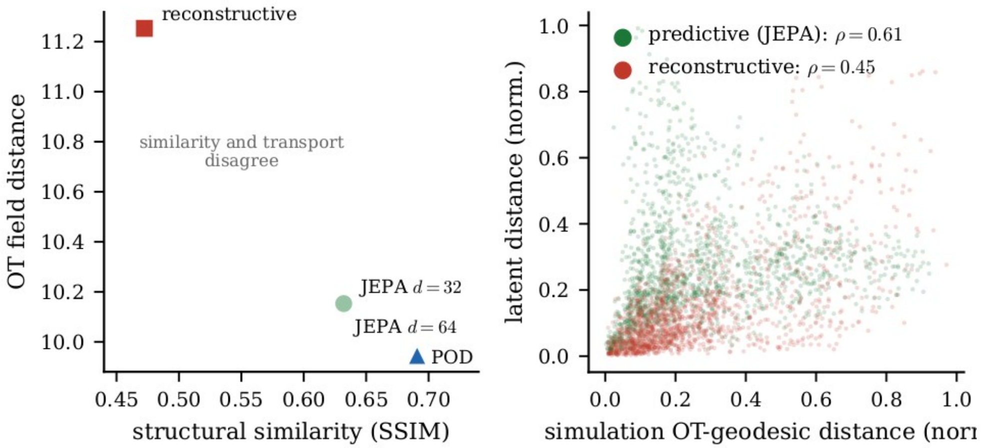
Backup slides

Supporting detail and ablations

Transport geometry (optimal transport)

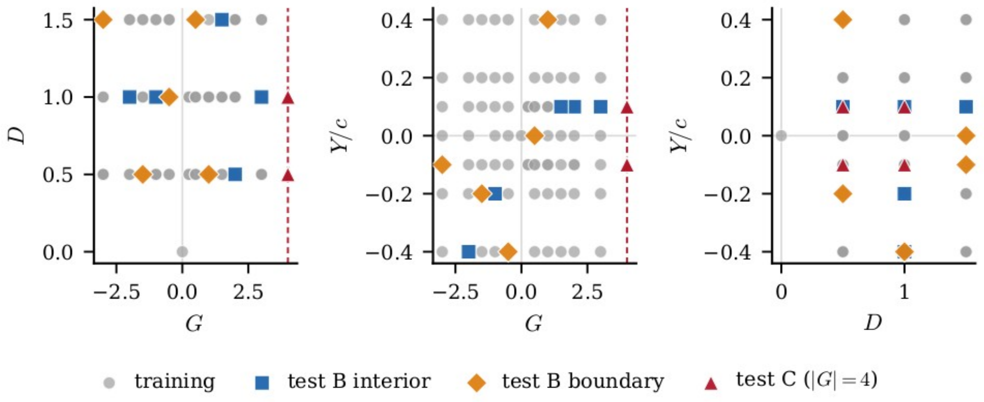
- **Why:** Euclidean / pixel distance ignores **where structures move**; control cares about advection of the LEV and shear layer
- **OT field distance** (after Tran, Yeh & Taira, JFM 2026): split vorticity into ± parts, transport each with **unbalanced Sinkhorn** OT and sum, the least work to rearrange one field into another
- We did **not** train the latent to match OT (that paper does); this is a **post-hoc test**: per encounter, Spearman-correlate the **latent** distance matrix with the **OT** matrix (n = 42)
- **Within-encounter:** the unconditioned JEPA **0.61** vs reconstructive **0.45**, order-preservation along the trajectory, **not** an isometry
- **Honest caveat:** *pooled across* encounters the alignment **does not hold**, so it is **trajectory-local**, not a global latent metric



Per-encounter latent-vs-OT distance alignment (Shepard). OT field distance after Tran, Yeh & Taira (JFM 2026). Within-encounter Spearman 0.61 vs 0.45; pooled it does not hold.

Dataset and protocol

- Partition **v2** (locked, sha256-anchored): 84 cases
- **226** train encounters · **42** test_b · **24** test_c (|G| = 4)
- ω_z pipeline v1: spatial mask + p99.99 clip + 3σ scale (train_std 3.55), no mean shift (preserves vorticity antisymmetry)
- Reporting: bootstrap n = 2000 CIs · 3 encoder seeds · 5-fold probe CV



Training envelope in (G, D, Y).

Matched-capacity: d = 32 vs d = 64

- **(Conditioned-model control; no unconditioned variant run.)** Halving the latent leaves the representation and **every mechanism diagnostic** intact
  - representational wake R^2 0.74 (d=32) vs 0.75 (d=64); drift ratio 0.86 vs 0.85; OT 0.61 vs 0.63
- Cost is **in-distribution forecast sharpness**: H = 16 wake closure 0.45 → 0.21
- Participation ratio ≈ 1.7, leading PC ≈ ¾ of variance → the effective dimension is a handful

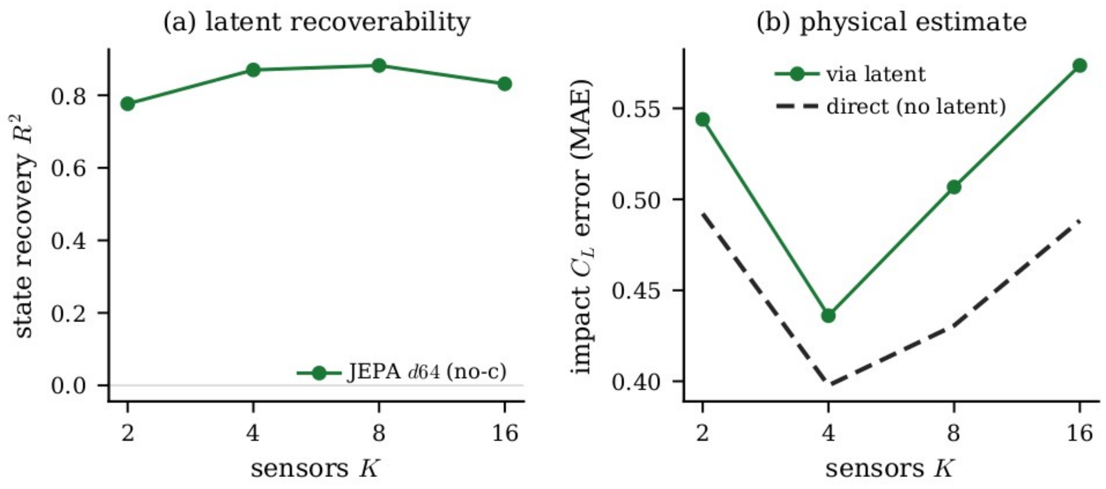
Seed variance

- **(Conditioned-model control; no unconditioned variant run.)** Three encoder seeds per cell; wake-closure means reported ± standard deviation
- The reconstructive CNN+ViT cell has **large seed variance (±0.27)**, consistent with the drift mechanism: without a forward-predictable geometry, closure is unstable across initialisations
- Predictive cells are tight (±0.03–0.06)

BACKUP

Parameter observability from the latent

- z → (G, D, Y) probe R^2 from the rolled-out latent, K = 8 pre-impact sensors
- test_b: **G 0.46 · D 0.80 · Y 0.10**, gust impulse and depth recoverable; cross-stream offset Y is marginal
- test_c (|G| = 4): negative, the 3D observability boundary of a single mid-plane slice



Gust parameters implicit in the latent on held-out cases.

SSIM convention

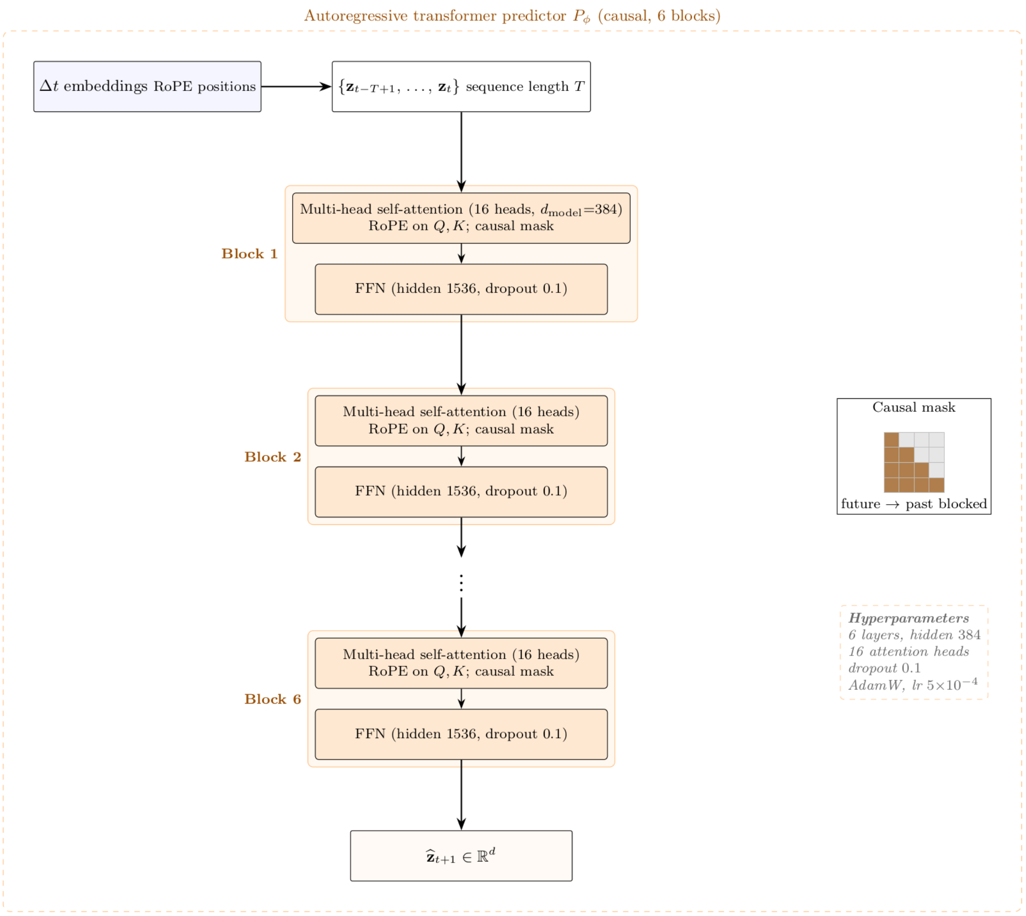
- Reconstruction SSIM uses the **Wang convention** (K1 = 0.01, K2 = 0.03) on pipeline-normalised ω
- Data range **L = 2 · global p99.9(|target|) ≈ 8.31** (split v2)
- Decoder test_a SSIM ≈ **0.73** at this convention

Parameter-only floor

- KRR-RBF on (G, D, Y) → observable at impact; train / test_b / test_c R^2
  - on train, (G, D, Y) interpolates C_L and C_D well (226 points in 3-D)
- On **test_b** the floor **collapses** across observables; on **test_c** it is negative for 5 of 6
- → JEPA's generalisation is **not** explained by the gust parameters alone

Model detail: encoder and predictor

- **Encoder** = a CNN stem (3 down-sampling stages) + a 6-layer Vision Transformer; the field becomes 288 patch tokens, and a learned [CLS] token is mapped by a small MLP (with BatchNorm) to the latent z (d = 32 / 64)
- **Predictor** = a 6-layer autoregressive transformer (hidden 384, 16 heads, dropout 0.1): it reads z(t) and emits the next latent z(t+1). **Unconditioned: no gust parameters enter the predictor**
- **Causal mask:** each step sees only past steps, never the future; **RoPE** encodes the frame ordering so the transformer knows what is earlier or later
- **Training:** match the next latent (teacher forcing) + **scheduled sampling** (feed the model its own predictions over an 8-step rollout, so it tolerates its own errors) + **SIGReg** anti-collapse (keeps z informative without a decoder); AdamW, bf16, 20k steps



Phase–amplitude reading

- Phase–amplitude decomposition of the encounter cycle in the predictive latent
- Connects the predictor to the **sensitivity-function control** designed for these flows
- An actuation channel is the natural next input for closed-loop control